

Working Version of Norm Experiments

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(intro, theory, setup, etc.)

3 Environment and Implementation

3.1 Environment

There are a set of states S in the environment, each state representing a different physical or social situation. For each state, there are a set of actions A that can be performed by agents. Thus, any combination (s, a) where $s \in S$ and $a \in A$ is a situation and a particular action that is applicable to the current situation.

At each timestep, there is a different state (situation) within the community to which agents must react.

3.2 Agents

There is a set of N agents, C , which represent the community. Agents are actors within the community; they perform actions in the different states as provided by the environment. Agents have different preferences for actions in different states, no matter whether they perform the action or another agent does, as determined by $q_i(s, a) : S \times A \rightarrow \mathbb{R}$ for each agent i . For example, agent i may appreciate another agent holding the door open for them as they pass ($q_i(s, a) > 0$), while another agent j may not mind at all ($q_j(s, a) = 0$).

Agents may not always be active or attentive of everything that goes on in a community. Community members may miss meetings, not attend events because of external commitments, or simply not participate as often as others. Agents are active within the community with a rate of R_i for each agent i .

Agents have three roles that they may take on within the community:

1. Actor: These agents do not follow any other agents. They perform actions in accordance to their own policy which **attempts to maximize their own utility**.
2. Follower: These agents follow another agent (an Actor or Influencer) which they have identified as a good role model. They perform actions by copying the policy of the agent whom they are following.
3. Influencer: These agents do not follow any other agents. They perform actions in accordance to their own policy which **attempts to maximize the utility of their followers**.

3.3 Opinion and Gossiping

Agents form their opinions of other agents in the community. Agents only interact in the community in one meaningful way: they share their opinions of other agents. For each pair of agents i and j , i keeps two opinion measures of j :

1. $P_{i,j}$, the **personal estimate** of j 's utility to i , representing the reward that i observes from j 's actions. This opinion is formed through a moving average which is updated each time i observes an action by j at timestep t using $P_{i,j} \leftarrow P_{i,j} \cdot (1 - \alpha) + \alpha \cdot U_{i,j}^{(t)}$, where $U_{i,j}^{(t)}$ is the utility that i observes from j at timestep t . Note that $U_{i,j}^{(t)} = 0$ if j is inactive, and $U_{i,j}^{(t)} = q_i(s^{(t)}, a_j^{(t)})$ if j is active, where $a_j^{(t)}$ is the action that j performs in timestep t . This moving average is not updated during timesteps where i is inactive.
2. $R_{i,j}$, the **reputation estimate** of j from the perspective of i , representing i 's estimate of the *community-at-large*'s opinion of j .

Agent i also keeps a **Status** estimate of itself, which is $S_i = f R_{i,i}$, where f is the number of followers i has. This is to estimate i 's total reputation within the community.

At the beginning, $R_{i,j}$ is equal to $P_{i,j}$. Any change in $P_{i,j}$ is also reflected in $R_{i,j}$, that is, $R_{i,j}^{(t+1)} = R_{i,j}^{(t)} + (P_{i,j}^{(t+1)} - P_{i,j}^{(t)})$. After the first timestep, the set of agents that are active (C_{active}) **gossip** with each other.

Agents gossip by averaging (through mean) their opinions of all other agents. That is, at each timestep t , for each active agent i in C_{active} , and for every agent j ,

$$R_{i,j}^{(t)} = \sum_{k \in C_{active}} \frac{1}{|C_{active}|} R_{k,j}^{(t)} \quad (1)$$

3.4 Community Actions

Agents adjust their involvement in the community, and switch roles based on their observed status and reputations of themselves and others within the community.

3.4.1 Rate Allocation

An agent i will change their level of involvement within the community λ_i based on the amount of utility that the community provides them. At any given timestep, there is an λ_i chance that i will be active, perform an action, update their P and R values, and gossip.

There is a utility from outside the community of b_0 , and an agent i allocates their rate linearly between b_0 and their own personal estimate, $P_{i,i}$. An agent i reallocates their rate at an average interval of δ timesteps, reallocating using the equation

$$\lambda_i = 1 - \exp\left(-\frac{P_{i,i}}{P_{i,i} + b_0}\right) \quad (2)$$

3.4.2 Role Switching

All agents begin as Actors, which do not follow other agents and optimize their behaviour to maximize their own (self) utility. An agent i re-evaluates their role at an average interval of δ timesteps. An agent i will switch to an

1. **Follower** if the agent j where $j = \arg \max_j R_{i,j}$ has it such that $R_{i,j} > P_{i,i}$. That is, the reputation of the top agent exceeds the utility that i can provide to itself.
2. **Influencer** if $f R_{i,i} > P_{i,i}$. That is, the total *perceived* reputation of i (its number of followers multiplied by the estimate of its own reputation) is greater than the utility i can provide to itself.
3. **Actor** if $P_{i,i} > f R_{i,i}$ and $P_{i,i} > R_{i,j}$ for all j . This means the agent gets the most utility from itself, and will continue to optimize for their own utility.

3.5 Scaling Modifiers

Within communities, there may be factors that affect how others interact in terms of their view on the value of reputation and status. For example, a community which gives numerous benefits to leaders or influencers (club president rights, financial incentives, etc.) will make the act of being an influencer for the betterment of the community more attractive. Additionally, in communities with high levels of interpersonal contact, agents may value their reputation in the eyes of others more than their own needs. These are embodied by γ and κ :

1. Reputation Modifier γ to augment the threshold at which an actor becomes a follower. An actor i will become a follower and follow agent j if $\gamma R_{i,j} > P_{i,i}$. This abstractly models benefits that come with following the norm that makes it more attractive towards agents.
2. Status Modifier κ augments the threshold at which an influencer optimizes for status instead of personal utility. An influencer i will optimize for status if $\kappa S_i > P_{i,i}$. This abstractly models benefits that come with status that make status more attractive towards agents.

3.6 Other Parameters

We take:

1. A rate and role update interval of 3000 steps.
2. An update factor of $3 \cdot 0.001$.
3. A gradient learning rate α of 0.0005.

4 Behavioural Convergence

In this section we investigate the convergence of agent reputations and behaviour. We look at a singular agent (agent 88) acting within a community with no other variables or factors. Within our environment, agents naturally only optimize for their own personal utility. Over time, the average personal utility for an individual agent towards a stable, high value, indicating that agents are effectively learning policies that maximize their individual reward functions.

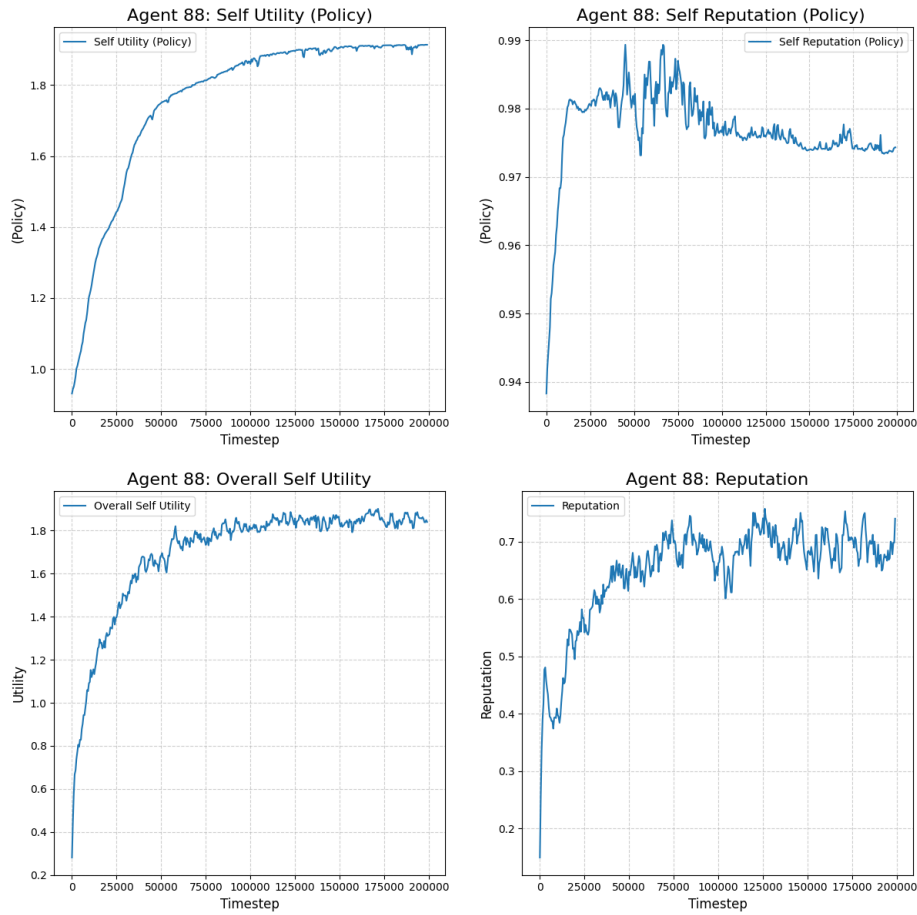


Figure 1: Top: the expected self-utility and reputation from Agent 88’s policy. Bottom: the true self-utility and reputation as within the environment. We can see that in terms of self-utility, Agent 88 has very close to convergence behaviour. Its reputation, however, does not converge.

This initial result validates the core learning mechanism of the agents. In the absence of social pressures to follow or lead, agents behave as classical rational or reinforcement learning agents, and we can see that they do so effectively.

Additionally, while it is possible for the reputation of an agent to converge like the personal utility, for many agents it is observed that their reputation, as perceived by the community, often fails to converge. Instead, it can exhibit high variance or persistent oscillations, even for agents who have maximized their own utility.

The instability of reputation can be attributed to the possible existence of multiple optimal policies for the agent’s personal

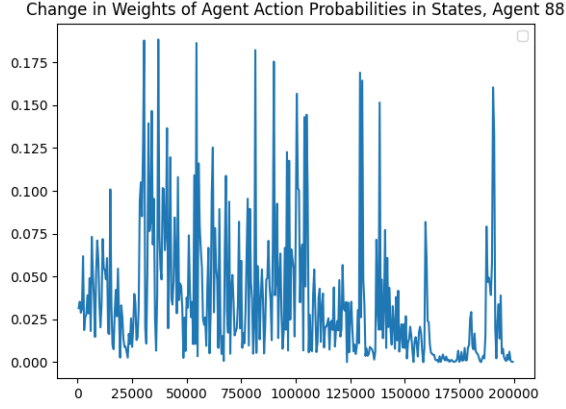


Figure 2: Absolute weight differences for the acting model of Agent 88.

utility. An agent's utility landscape may contain multiple, equally optimal policies. For example, an agent i might discover two different actions, A_1 and A_2 , that both yield its maximum personal utility. Agent j is therefore indifferent and may alternate between A_1 and A_2 . However, from the perspective of an observing agent i , A_1 might generate high utility ($U(i, j)$ is high) while A_2 might generate low or even negative utility ($U(i, j)$ is low). As agent i switches between its equally-good-for-it policies, its reputation will fluctuate wildly, making it appear erratic and unpredictable to the community.

5 Reputation Scaling

In this section we study the effects of γ on the community. γ controls the number of agents which participate in the norm under the influencer. A high γ value would lead to more agents (up to $n - 1$) agents following the influencer. A lower γ value would lead to less or no agents following the influencer. We varied γ to observe the final follower counts depending on different values, with the goal to see when it is sufficient for a single agent to gain all the followers in the community. In these experiments, there are no influencer benefits of any kind ($\kappa = 0$) and agents only optimize for their own personal utility.

Table 1: Highest Follower Count for Top Agent with Different γ values (with $\kappa = 0$), Static Switching

γ	0	1	1.25	1.5	1.75	2	3	5
Followers	0	2	9	37	82	92	99	99

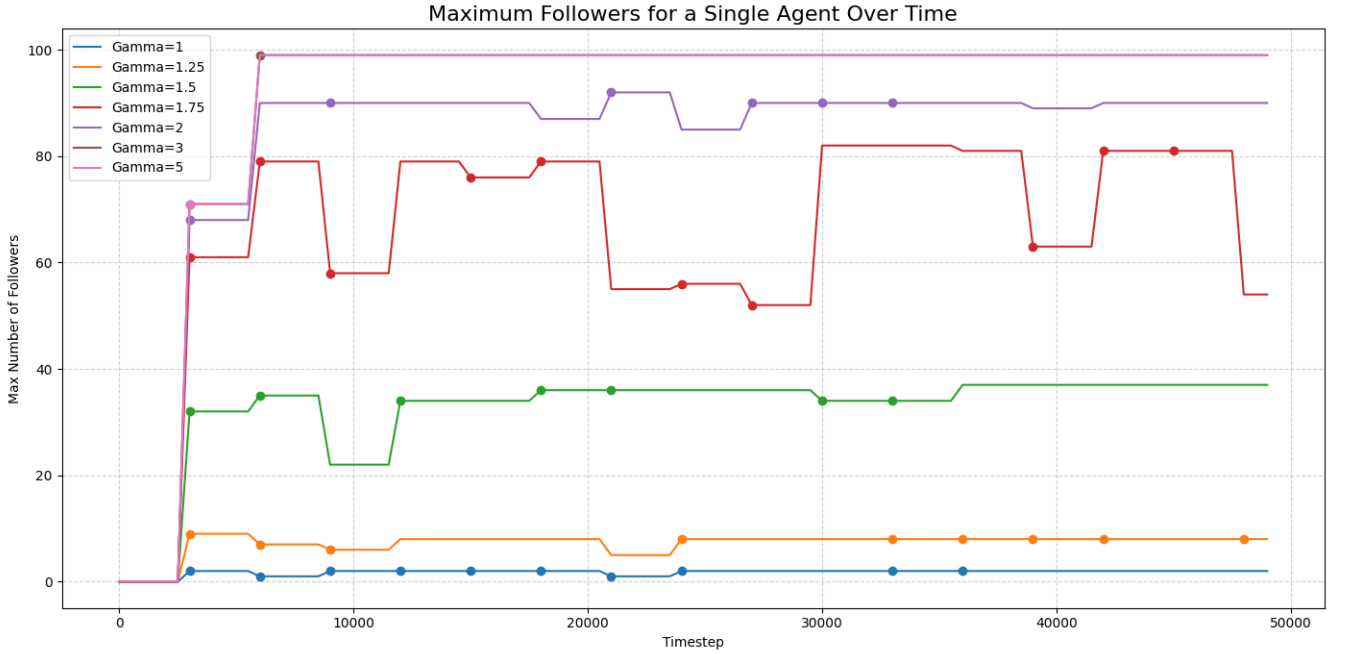


Figure 3: Summary of experiments for γ for **static**. Number of Followers within each community. The higher γ is chosen to be, the higher number of followers (and thus the number of participants in the eventual norm) will be. Blips show the top influencer switching.

The number of agents that participate in the norm increases from 0 at $\gamma = 0$ to 99 at $\gamma = 3$. As shown in the table, increasing γ has a dramatic, non-linear effect on the social structure. At $\gamma = 1$, social influence is minimal, resulting in only a handful of followers for the most reputable agent. However, as γ crosses a threshold around 1.5, the system undergoes a rapid phase transition. The size of the dominant agent's follower base explodes, and by $\gamma = 2$, the system has effectively consolidated into a unipolar structure, with a single leader being followed by nearly the entire population (92 out of 99 possible agents).

A higher γ also benefits the norm convergence by separating the reputation of agents and pruning the competition for top agent. Because agents are not allowed to follow other followers, the moment a contender agent k is "captured" by another, agent j , it is removed from the pool of potential leaders. All the agents that might have followed agent k are now forced to switch their influencer, possible to agent j .

We found that $\gamma = 3$ is a sufficient γ value to ensure that the entire community converges under a single influencer under any conditions, and thus will use it in subsequent experiments.

Table 2: Highest Follower Count for Top Agent with Different γ values (with $\kappa = 0$), Asynchronous Switching

γ	0	1	1.25	1.5	1.75	2	3	5
Followers	0	3	9	34	79	92	99	99

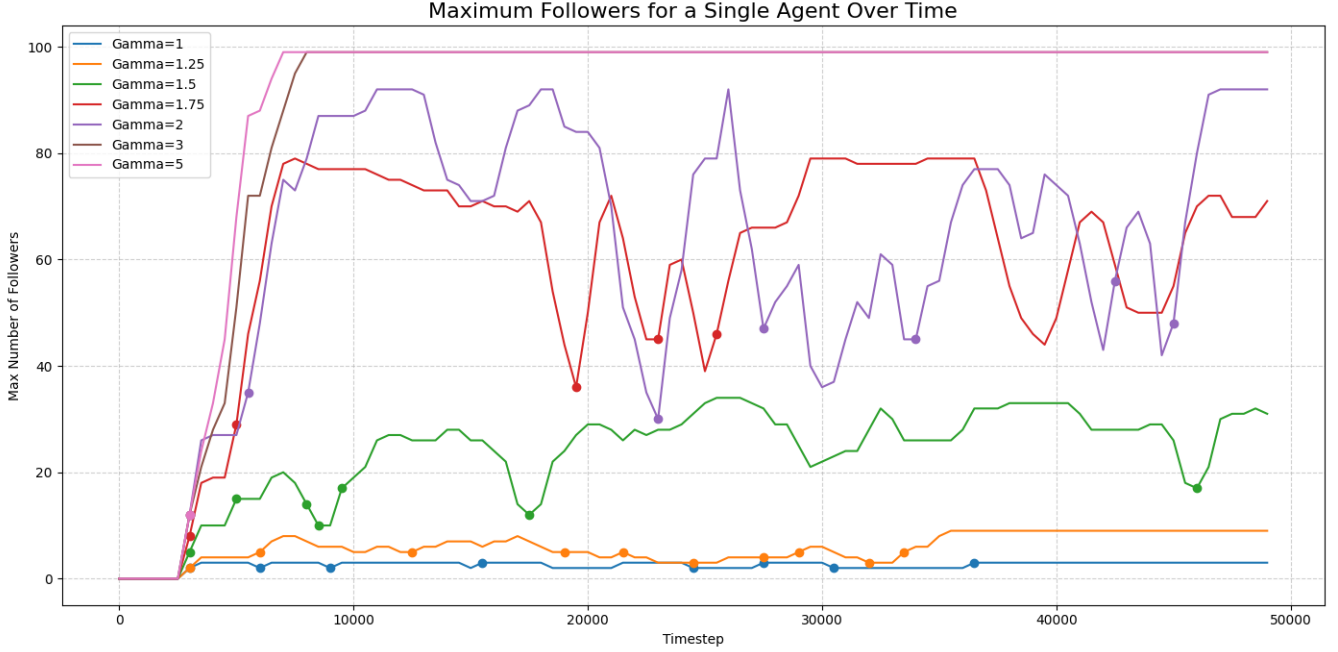


Figure 4: Summary of experiments for γ for **asynchronous**. Number of Followers within each community. The higher γ is chosen to be, the higher number of followers (and thus the number of participants in the eventual norm) will be. Blips show the top influencer switching.

6 Status Scaling

Now that we have established the effect of γ on ensuring the emergence of a common norm, we will study how status encourages that the common norm is an optimal norm. In this section we study the effects of κ on the community. κ controls whether the influencer will optimize for personal utility or for status. We will increase κ (to represent the benefits of the influencer) and see when the influencer begins to optimize for status.

6.1 Optimizing Objective

The most immediate result to be observed from status scaling is that the top agent will begin to optimize for status once κ exceeds a certain value, as then $\kappa \cdot S > P_{i,i}$. For values of $\kappa \leq 0.01$, the influencer continues to behave as a self-interested actor, optimizing for its personal utility $P_{i,i}$. However, for all tested values of $\kappa \geq 0.05$, the agent switches its strategy to become an influencer which is optimizing for status S .

Table 3: Influencer focus with different κ values (with $\gamma = 3$), synchronous

κ	0.001	0.01	0.02	0.05	0.1	0.5	1
Optimizing for Status?	✗	✗	✗	✓	✓	✓	✓

Table 4: Influencer focus with different κ values (with $\gamma = 3$), asynchronous

κ	0.001	0.01	0.02	0.05	0.1	0.5	1
Optimizing for Status?	✗	✗	✗	✓	✓	✓	✓

6.2 Optimal Social Welfare

The norm policy value is defined as the expected reward for the community from an action by the top agent at any given timestep. The leader's shift in motivation has a direct and significant positive impact on the community. The final expected reward of the converged norm increases from 775 to ≈ 923 once the agent begins optimizing for status. Once this threshold is crossed, further increases in κ do not yield additional community benefits.

Table 5: Highest norm policy value with different κ values (with $\gamma = 3$)

κ	0.01	0.02	0.05	0.1	0.5	1	5
Final Policy Value	775	775	918	920	922	925	923

6.3 Convergence

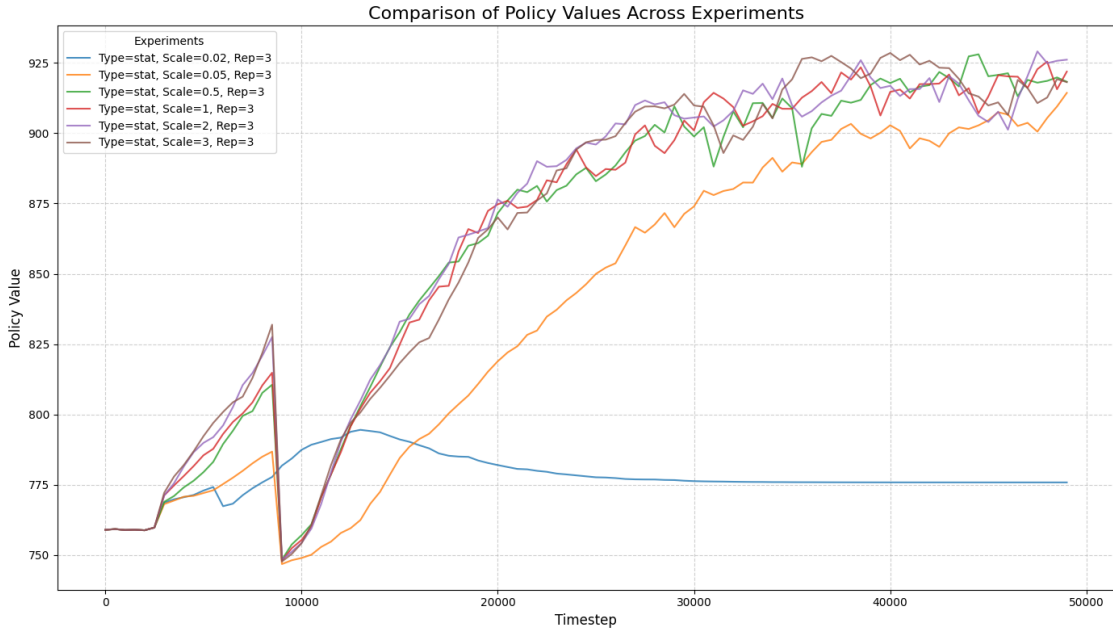


Figure 5: Summary of experiments for κ . Policy Progression Graphs of the Top Agent in each experimental run.

Agents with $\kappa \geq 0.05$ optimize for status, while agents with $\kappa \leq 0.02$ optimize for personal utility. There is a drop in the reputation between $\kappa = 0.05$ and $\kappa = 0.02$ that is much larger than the differences between other values on either side.

As a higher κ directly increases the value of S , this directly impacts the top agent's rate within the community, and thus has a higher probability of being active in any given timestep. As an agent can only learn and adapt its policy when it is active. A higher activity rate r_i means the leader gets more "turns" to take actions, observe the outcomes (via the reputation feedback from its followers), and update its policy. This increased frequency of action and feedback means the top agent will converge faster with a higher κ .

7 Variance

7.1 Unimodal

In this section we study the behaviour of the community when we change the reward function. We sampled the agent reward for any state and any action using a normal distribution with $\mu = 0.5$ and $\sigma = 0.5$. Note that the threshold set here is 99, so agents under such a threshold in followers will automatically not be optimizing for status.

In the previous sections, we established that high social pressure ($\gamma=3$) can induce consolidation around a single leader. However, those simulations were conducted under conditions of relatively low preference heterogeneity ($\sigma = 0.5$). In this section, we investigate how the diversity of agent reward functions, controlled by the variance parameter σ , impacts the stability of this centralized social structure. We maintain $\gamma = 3$ and $\kappa = 0$.

When σ is low, agent reward functions are highly correlated. A policy that is good for the leader is also likely to be "good enough" for most other agents. The utility a follower i gets from copying the leader l , $P_{i,l}$, is high. The leader's community reputation is therefore also high, easily overcoming the $P_{i,i}$ of most agents when amplified by $\gamma = 3$.

When σ is high, agent reward functions diverge significantly. A policy that is optimal for the leader would provide far less utility for individual agents, especially as compared to $P_{i,i}$, which the agent itself is able to optimize.

Table 6: Maximum Followers for Any Single Agent, $\kappa = 0$

σ	0.1	0.25	0.5	0.75	1	2
Followers	99	96	63	40	18	12

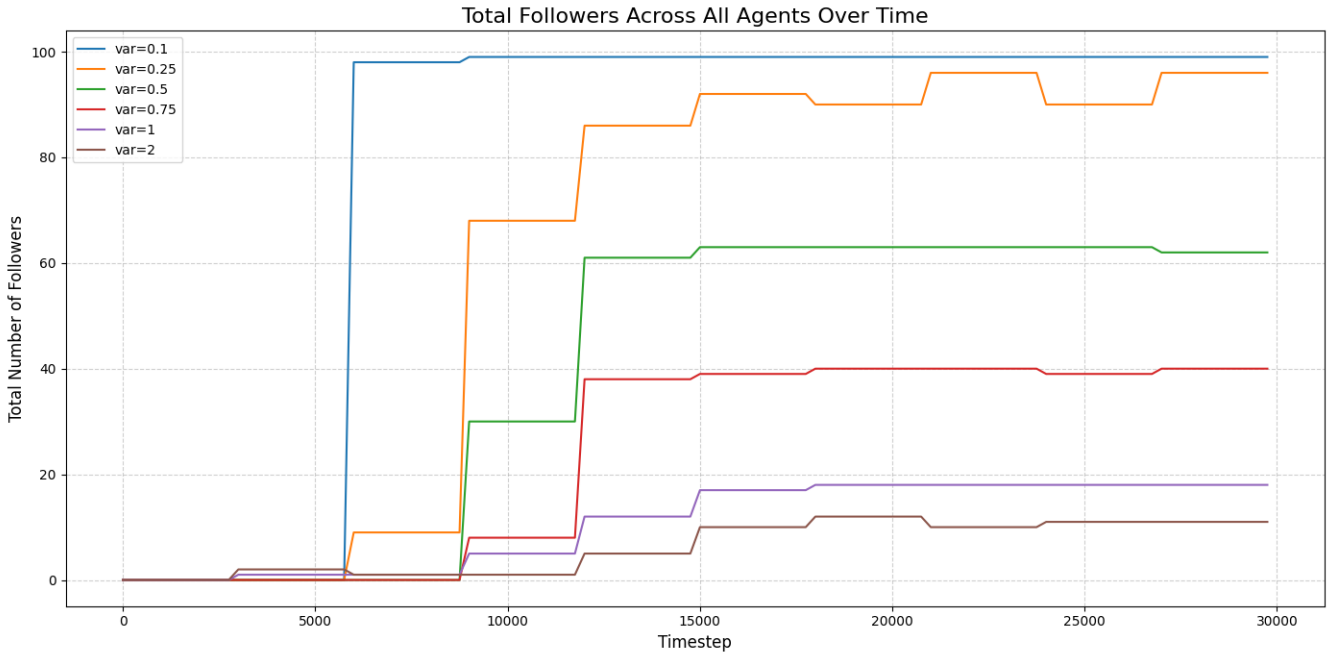


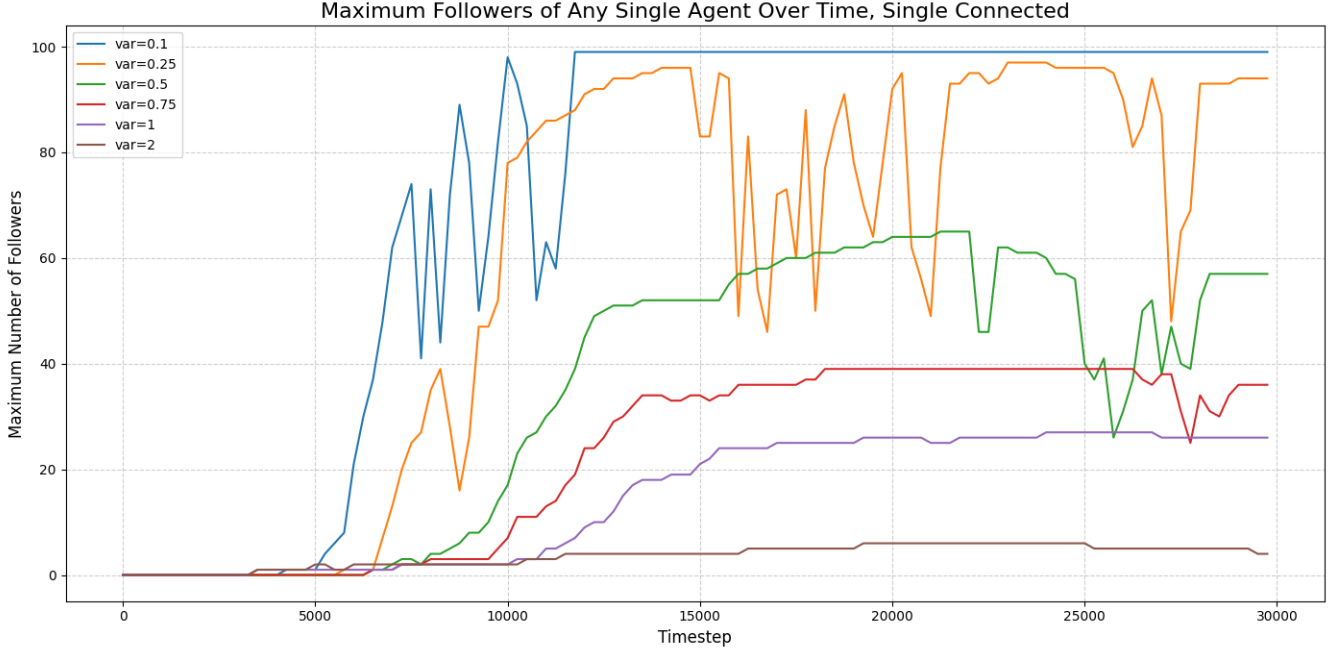
Figure 6: Summary of experiments for σ with $\kappa = 0$. Total Followers of all agents over time.

σ acts as a direct counter-force to γ . While γ pushes for conformity, σ pulls for individuality by making agents' goals more dissimilar.

This shows us that as heterogeneity increases, the power of reputation (γ) and leadership incentives (κ) is not absolute. A single leader cannot unite a population of agents with fundamentally orthogonal goals. An increase in γ or κ will be needed to compensate.

Table 7: Maximum Followers for Any Single Agent, $\kappa = 0$

σ	0.1	0.25	0.5	0.75	1	2
Followers	99	94	57	36	24	6

Figure 7: Summary of experiments for σ with $\kappa = 0$. Total Followers of all agents over time.

7.2 Bimodal

Table 8: Maximum Followers for Any Single Agent, Synchronous, $\sigma = 0.6$

Separation	1.25	1.5	1.6	1.65	1.7	1.75	2	3	5
Followers	99	99	99	45	31	28	14	0	0

When the ideological separation between groups is low (Separation ≤ 1.6), the strong social pressure ($\gamma = 3$) is sufficient to maintain a unified social hierarchy with a single leader commanding near-total followership. However, the moment the separation crosses a critical threshold (between 1.6 and 1.65), the system experiences a catastrophic collapse in consensus, with the dominant leader’s follower count plummeting by more than half. Further increases in separation lead to the complete dissolution of any cross-group hierarchy.

Let us assume that the leader l is in Group A_1 . When separation is low, the negative value for A_2 is not catastrophically bad. For a A_2 agent, following the A_1 leader is ideologically unappealing but may still be a better option than being an actor, especially when the social pressure ($\gamma = 3$) to conform is immense. The leader’s overall reputation, while slightly dragged down by A_2 ’s lukewarm reviews, remains high enough to sustain the hierarchy. The cost of non-conformity is greater than the cost of compromise.

When separation is high, the value of the policy of the leader l in the eyes of A_2 is very low. The policy adopted by l is now actively detrimental to the interests of every agent in A_2 . This causes the reputation of l to drop significantly, causing the agents in A_2 to stop following and instead be an independent actor.

Table 9: Maximum Followers for Any Single Agent, Asynchronous, $\sigma = 0.6$

Separation	1.25	1.3	1.4	1.5	1.75	2	3
Followers	99	99	99	19	8	3	0

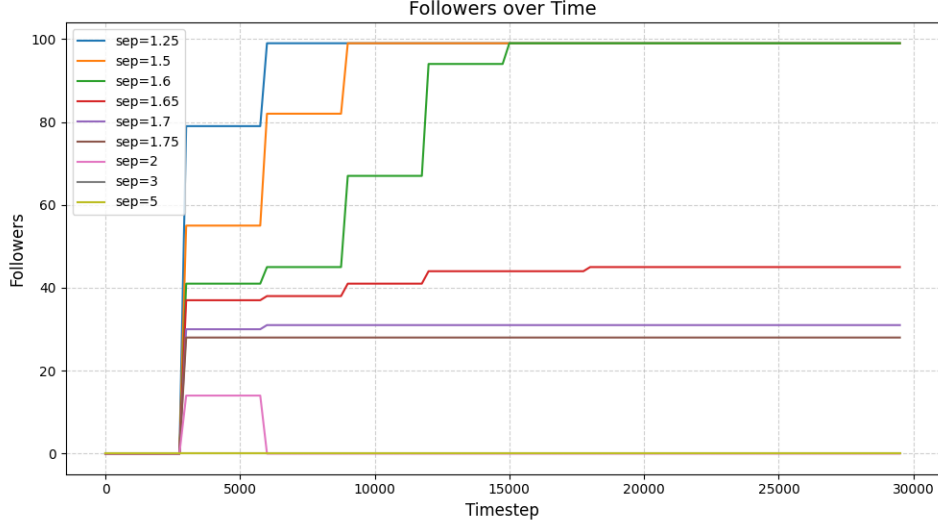


Figure 8: Summary of experiments for separation with $\kappa = 0.1$. Follower Progression Graphs of the Top Agent in each experimental run. Synchronous.

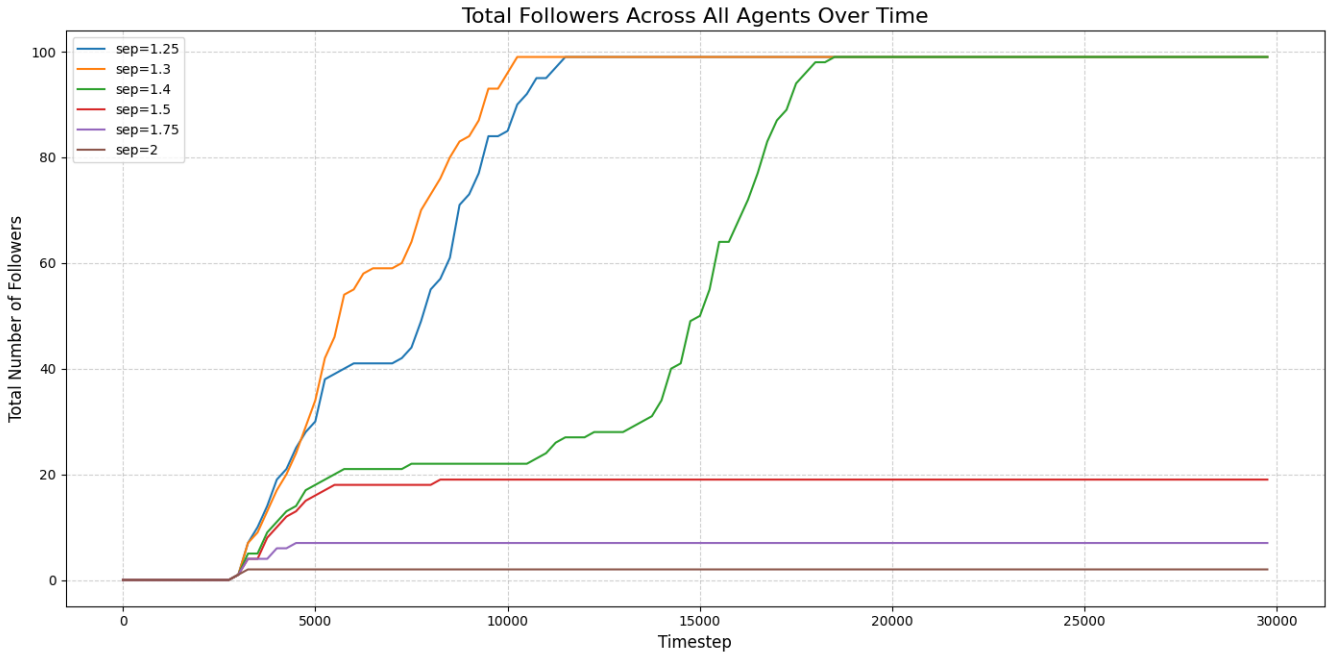


Figure 9: Summary of experiments for separation with $\kappa = 0.1$. Follower Progression Graphs of the Top Agent in each experimental run. Asynchronous.

8 Bimodal: Polarized Communities

We use the bimodal assumptions of two sets A_1 and A_2 which have conflicting reward functions. We let a probability ϵ be the probability that an agent from A_1 and an agent from A_2 will interact during gossiping. The probability that two agents from A_1 or two agents from A_2 share reputations is still 1.

Table 10: Convergence Times of Communities based on ϵ , Fully Connected

ϵ	1	0.5	0.1	0.01	0.001	0.0001	0
Timestep	6000	6000	9000	9000	12000	27000	Never

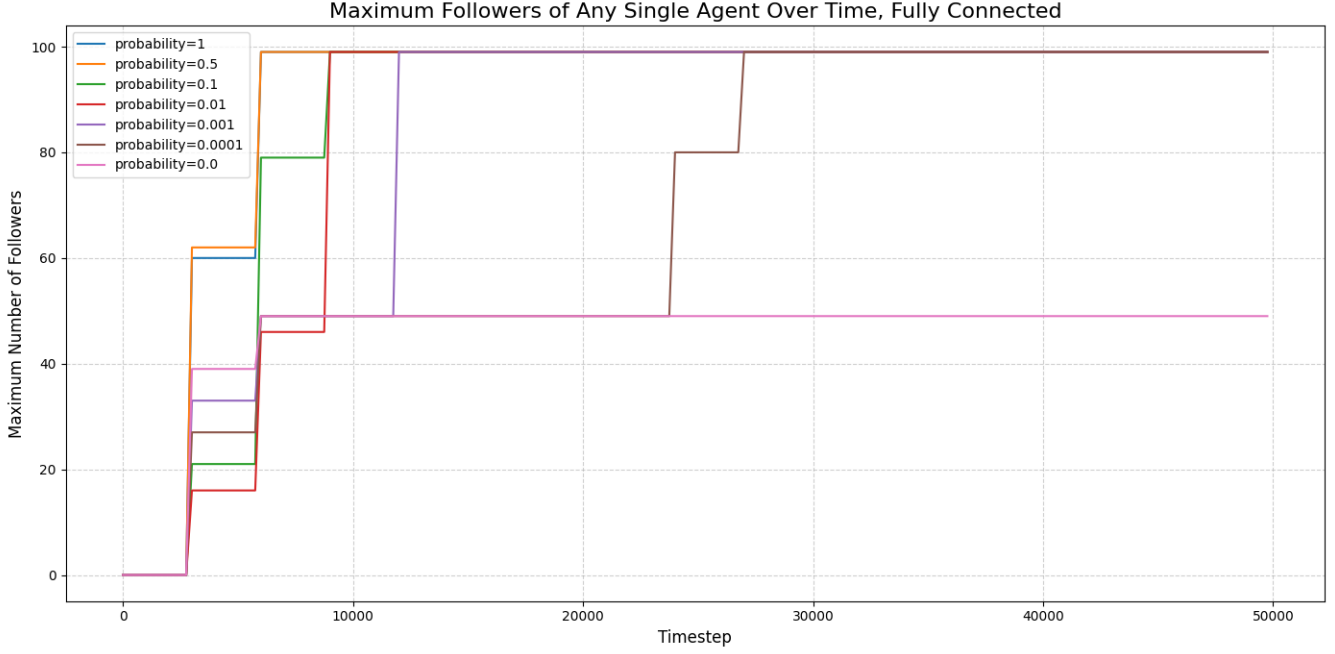


Figure 10: Summary of experiments for the fully connected ϵ probabilities. Follower progression graphs of the top agent in each experimental run.

8.1 Fully Connected

8.2 Single Connection

We now limit the connection between groups A_1 and A_2 to a single agent connection. We choose agent 1 from A_1 and agent 2 from A_2 .

Table 11: Convergence Times of Communities based on ϵ , Singular Connection

ϵ	1	0.5	0.1	0.01	0
Timestep	12000	12000	15000	Never	Never

We can see that no connection causes the community to (expectedly) fracture into two separate groups with two leaders. However, we can see that for a fully connected community, the community always eventually converges to a single agent. On the other hand, a single connection with a low enough ϵ may not be strong enough to make a difference in the two communities.

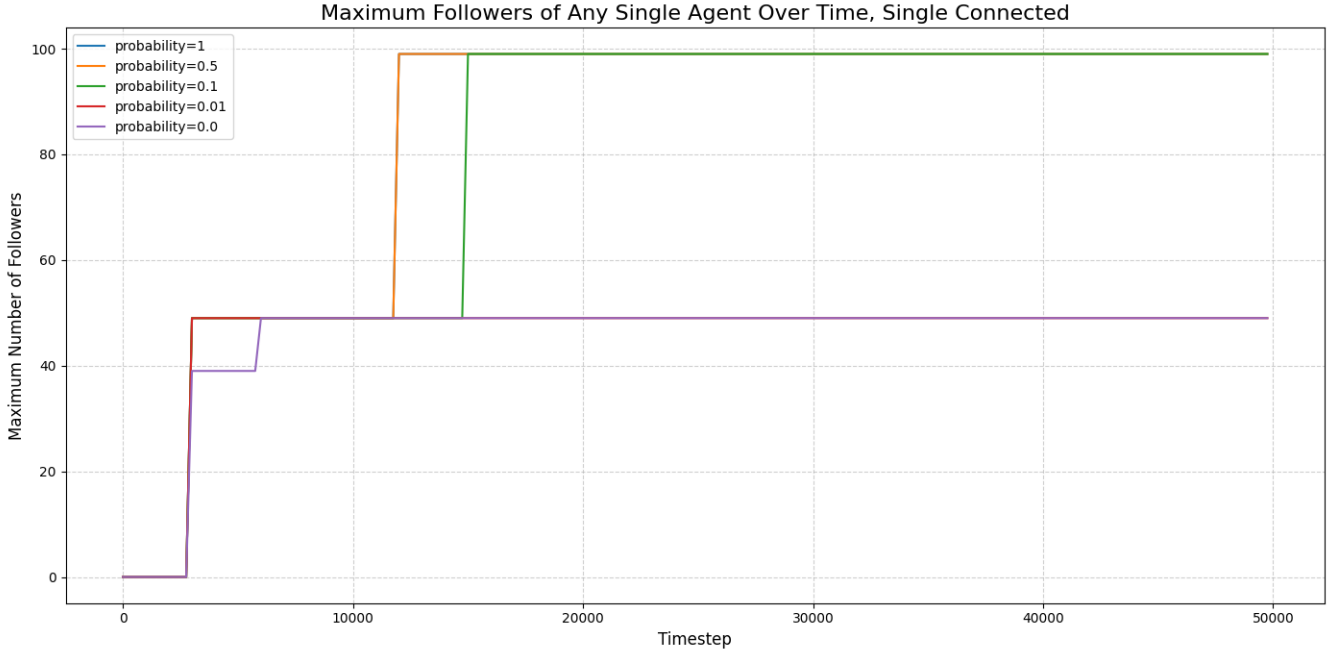


Figure 11: Summary of experiments for the singularly connected ϵ probabilities. Follower progression graphs of the top agent in each experimental run.

8.3 Dynamic Binary Connections

8.3.1 Similarity Distance

We explore a scenario where we employ a dynamic and binary communication style for gossiping. For any two agents i and j , we introduce the *similarity distance* score $Sim_{i,j}$, where $Sim_{i,j} = Sim_{j,i}$. This is a moving average based on i and j 's observations of each other's reaction to actions.

At each timestep in state s , for every action a taken by the active agent group, for all pairs of agents i and j , if i and j are both active, then

$$Sim_{i,j} \leftarrow Sim_{i,j} \cdot (1 - \alpha) + (q_i(s, a) - q_j(s, a))^2$$

We choose an arbitrary cutoff of 3, though in part because it is the smallest integer larger than the maximum reward, 2.25. That is, for an agent i , if $Sim_{i,j} < 3$, then i will gossip with j . Otherwise, i will not gossip with j .

8.3.2 Reputational ϵ

We also make it so that agents no longer choose explicitly the agent with the highest reputation when choosing a follower. Instead, an agents chooses

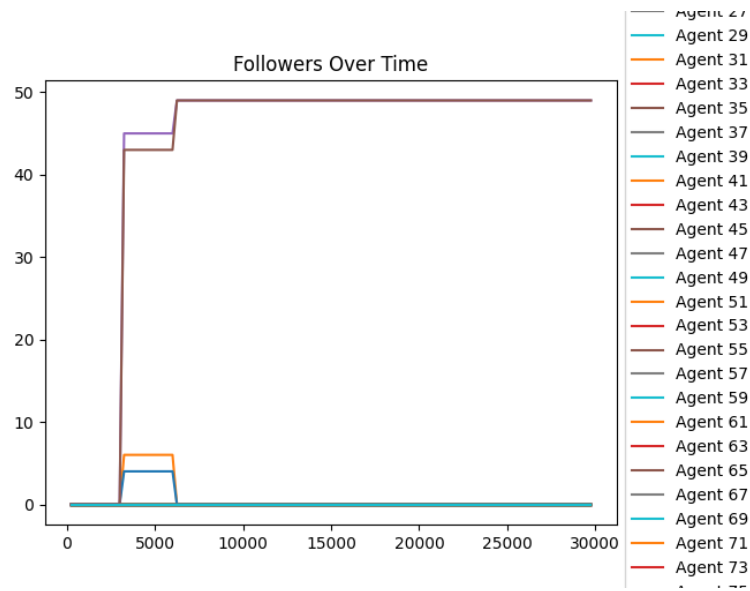


Figure 12: Follower progression for the Dynamic Binary connections (**Synchronous**). Notice we have two influencers, both at 49 agents.

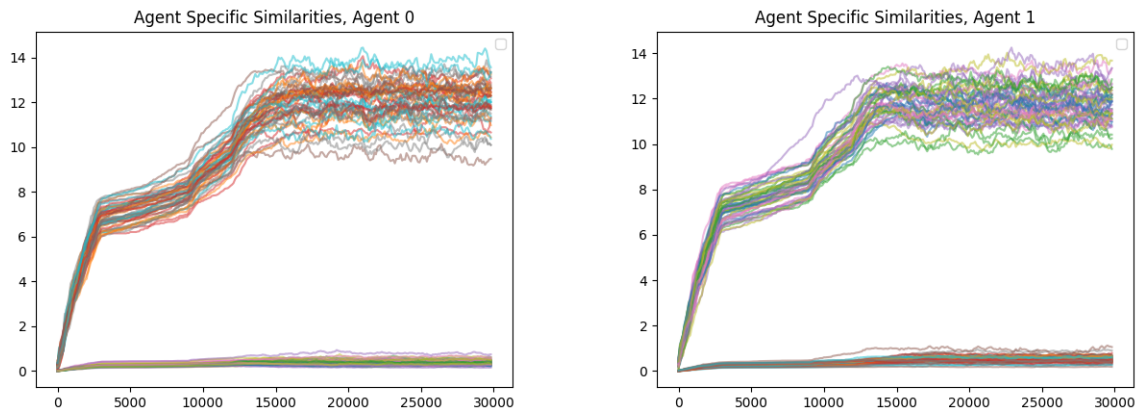


Figure 13: Similarity distance scores for all agents when compared with agent 0 (left, group A_1) and agent 1 (right, group A_2). Note that the other group of agents have very high distances, while the agent's own group has distances consistently below the threshold (3).

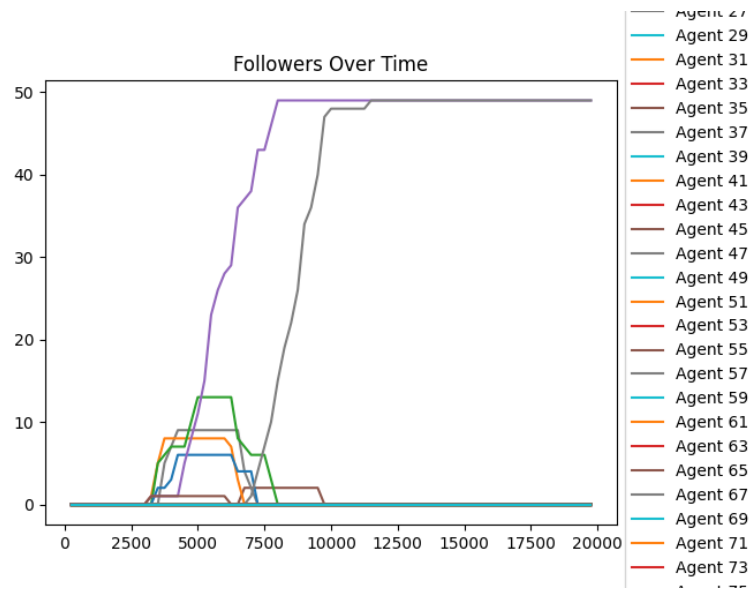


Figure 14: Follower progression for the Dynamic Binary connections (**Asynchronous**). Notice we have two influencers, both at 49 agents.

9 Appendix

9.1 Behavioural Convergence

Some agents do however converge (unlike Agent 88):

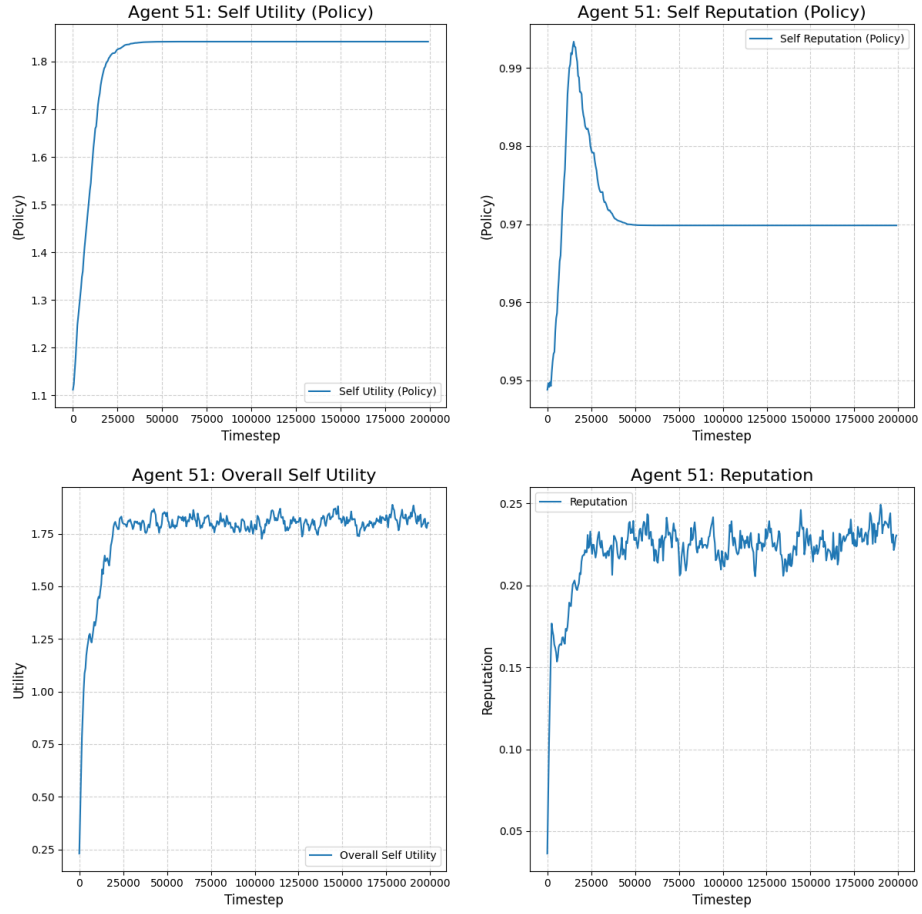


Figure 15: Top: the expected self-utility and reputation from Agent 51's policy. Bottom: the true self-utility and reputation as within the environment.

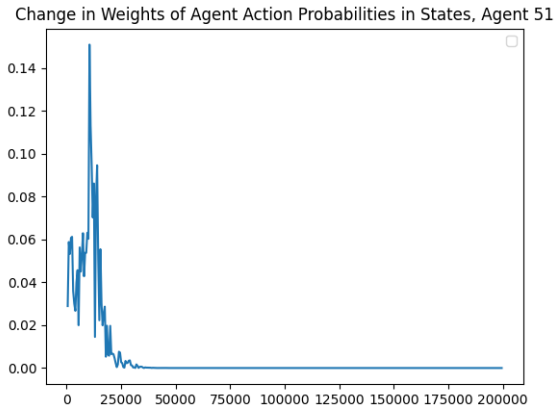


Figure 16: Absolute weight differences for the acting model of Agent 51.

9.2 Reputation Scaling

9.2.1 Static

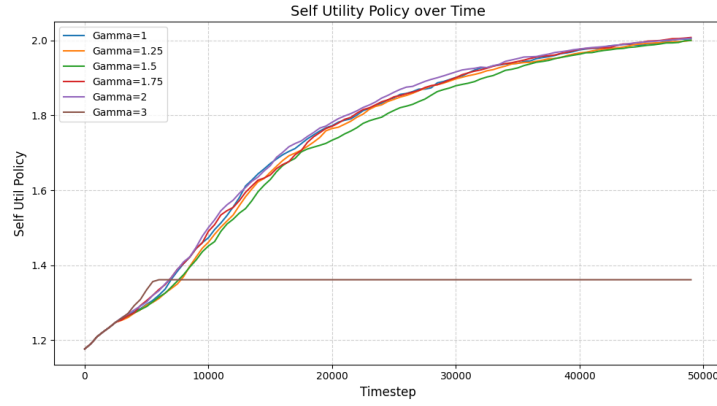


Figure 17: Agent 55 Policy-level Personal Utility. Note that Agent 55 is not the final influencer in $\gamma = 3$.

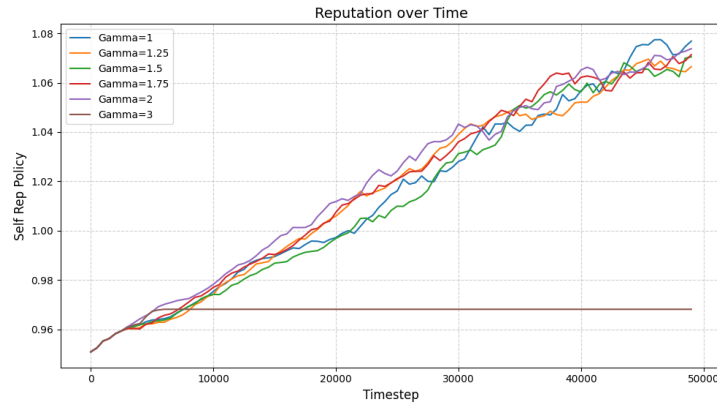


Figure 18: Agent 55 Policy-level Reputation. Note that Agent 55 is not the final influencer in $\gamma = 3$.

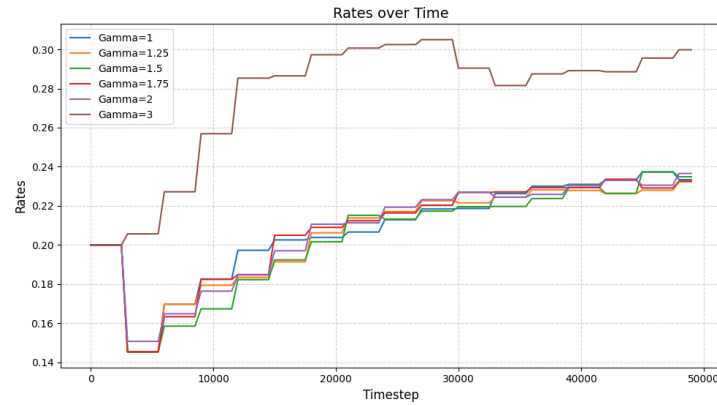


Figure 19: Agent 55 Policy-level Rates. Note that Agent 55 is not the final influencer in $\gamma = 3$.

9.2.2 Asynchronous

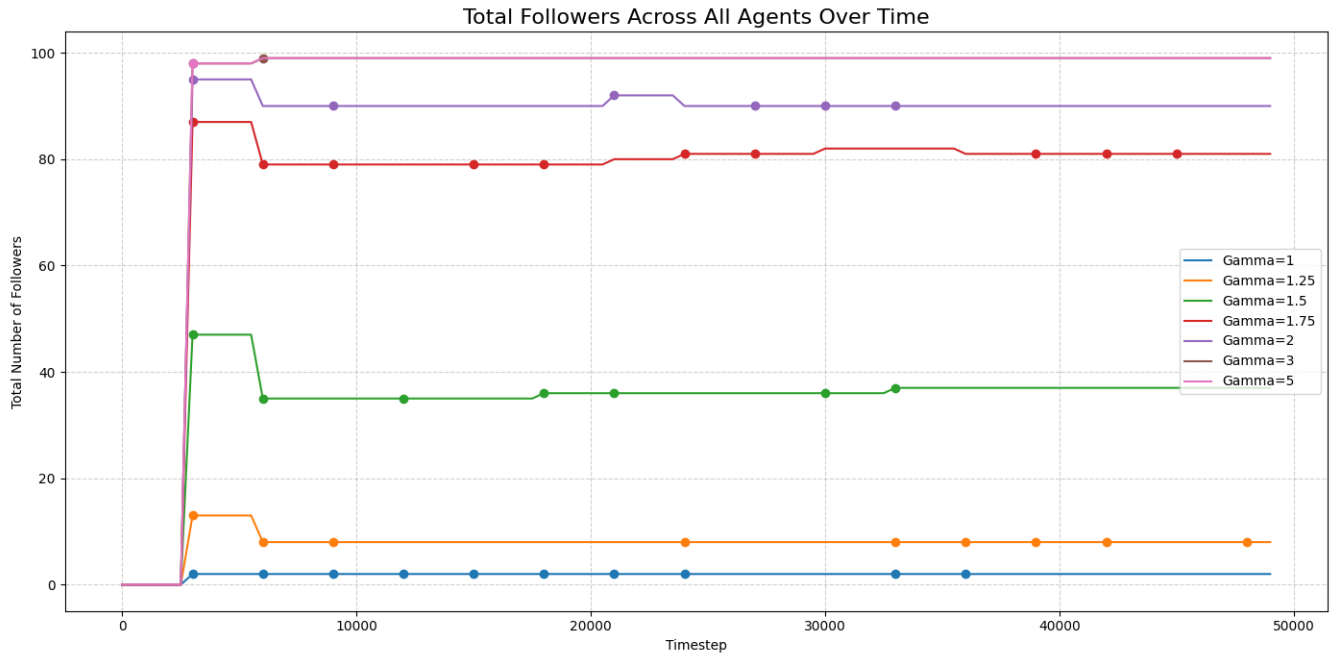


Figure 20: Summary of experiments for γ for **static**. Total number of followers overall.

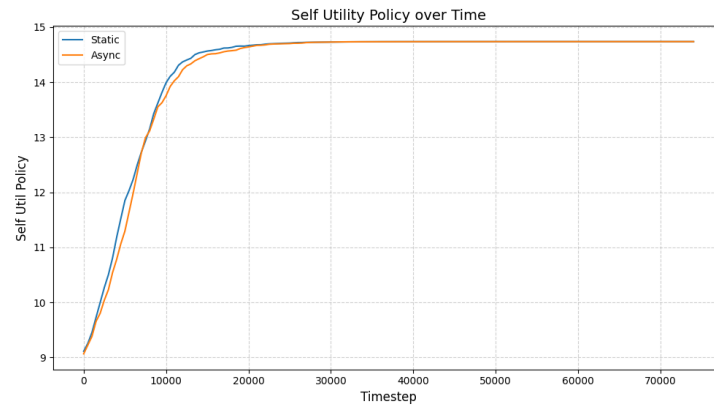


Figure 21: Personal Utilities of Agent 51 for $\gamma = 2$ for Asynchronous and Static.

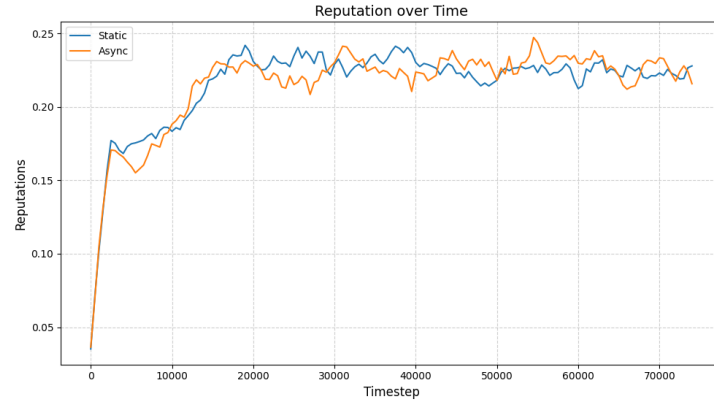


Figure 22: Reputations of Agent 51 for $\gamma = 2$ for Asynchronous and Static.

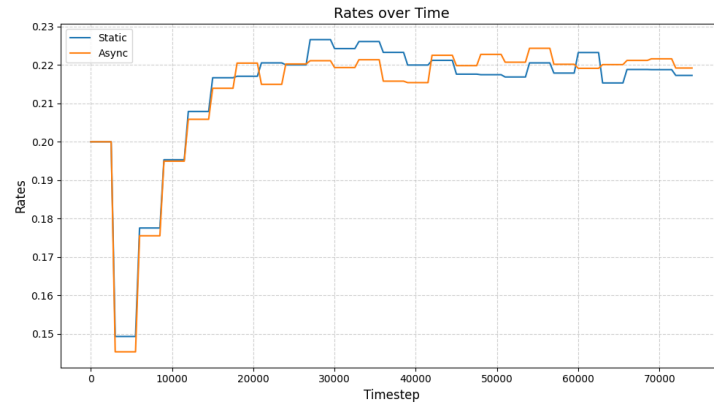


Figure 23: Rates of Agent 51 for $\gamma = 2$ for Asynchronous and Static .

9.3 Status Scaling

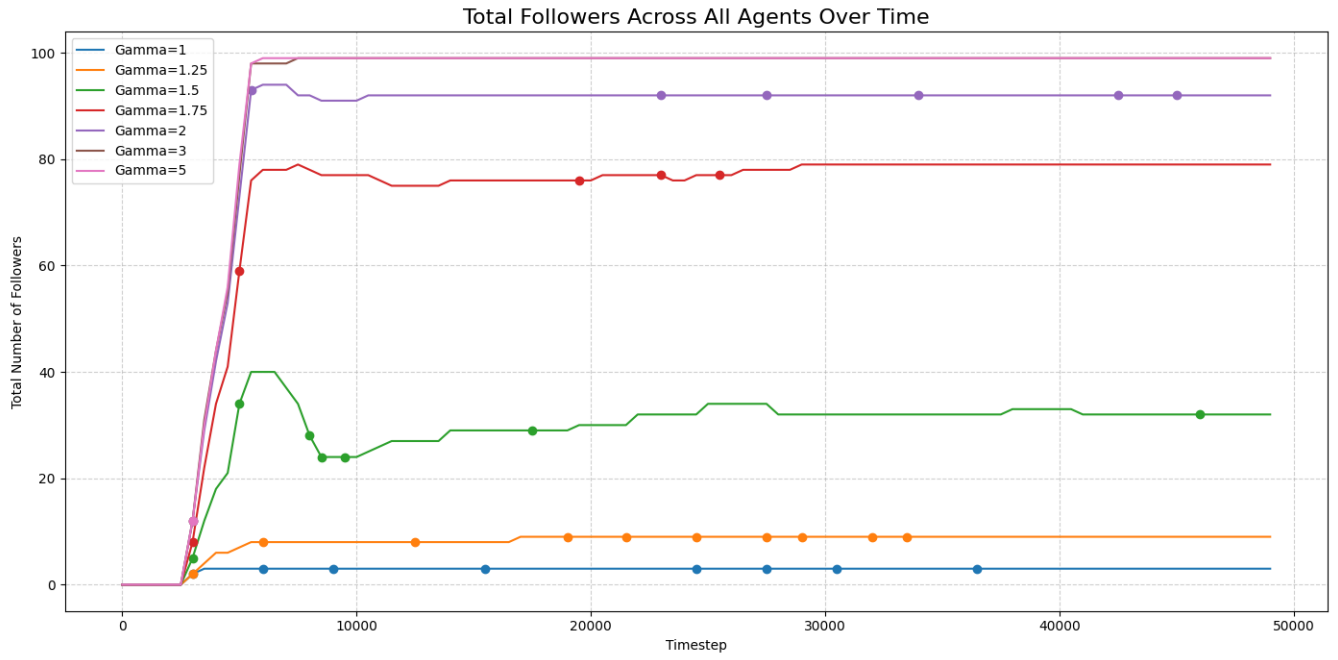


Figure 24: Summary of experiments for γ for **async**. Total number of followers overall.

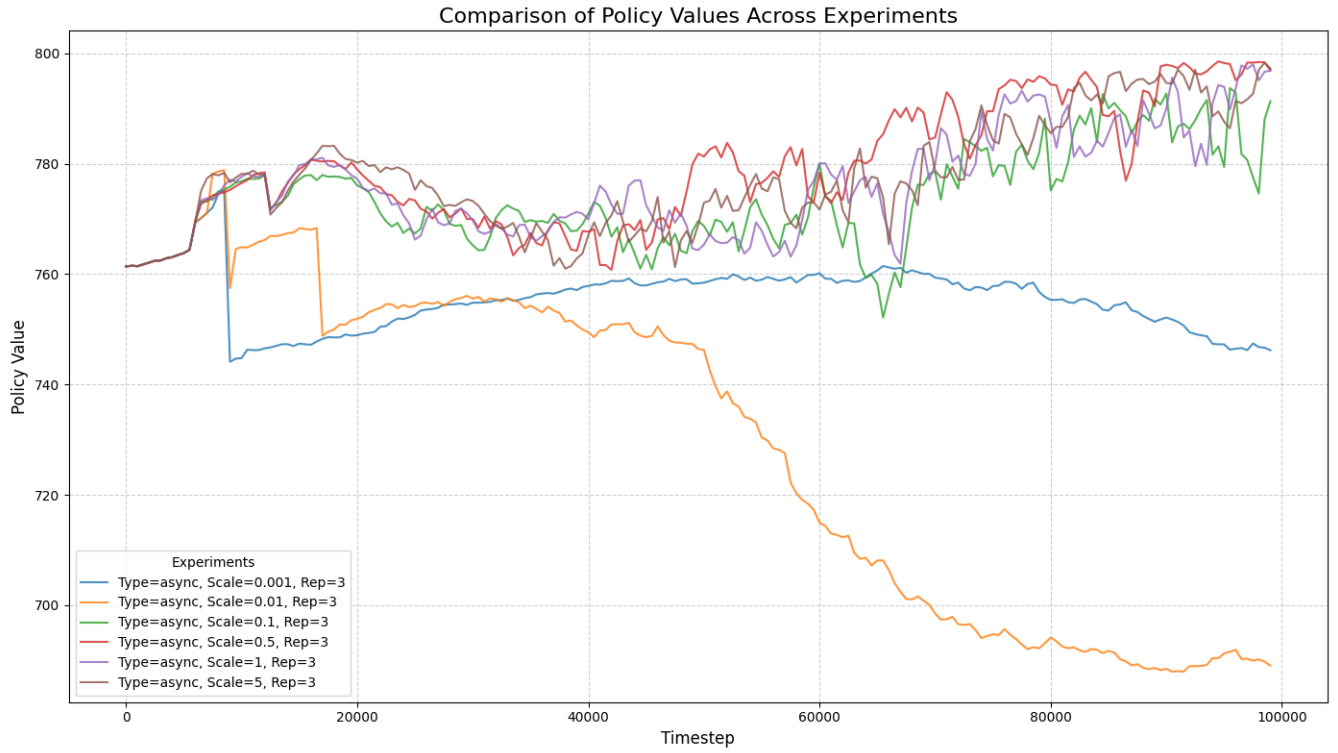


Figure 25: Summary of experiments for κ for **async**. Note that the top agent for 0.001, 0.01, and $\kappa > 0.1$ are all different (i.e. three different agents).

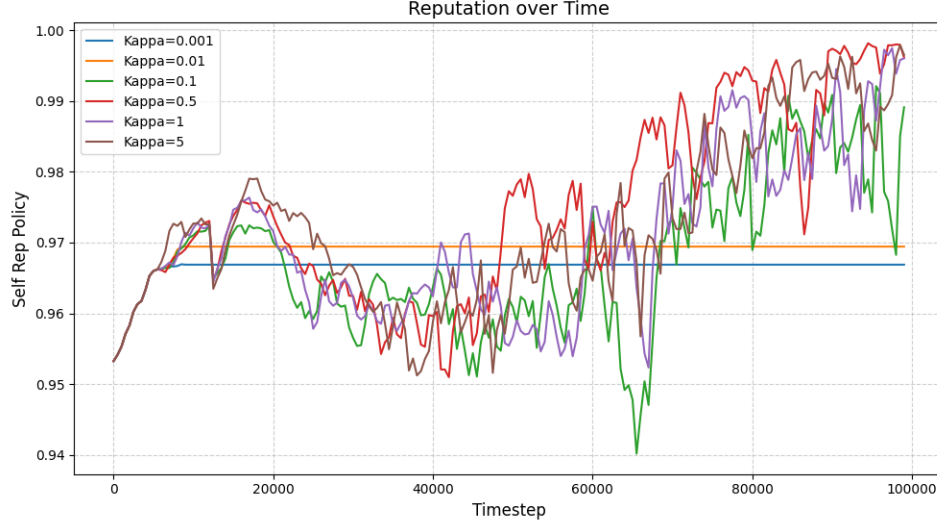


Figure 26: Summary of experiments for κ for **async** for agent 97's policy across the community.

9.4 Influencer Threshold

In this section we will study the effects of ρ on the community. ρ controls the threshold, in terms of the number of followers, that is required for an influencer to begin to optimize for status.

The most direct consequence of this is that a lower ρ creates the conditions for an influencer to optimize for status earlier than a higher ρ . An influencer only optimizes for status when both $\kappa S_i > P_{i,i}$ and $f_i > \rho$.

As agents only provide feedback to an influencer if they are following (and the influencer only receives this feedback if they are optimizing for status), the influencer will optimize towards maximizing the status from *their followers*, which may not be representative of the community-at-large. An agent i optimizing towards this suboptimal norm would cause its reputation among the community to decrease.

Hypothesis: A low ρ will cause influencers to (prematurely) optimize to a suboptimal norm because they are learning with data from a limited subset of the community, which will cause their reputation to fall. This would lead to a lowered number of followers as another agent may become the agent with the highest reputation.

This can be observed by plotting the reputation of the agent i and the number of followers of agent i . We can confirm this by creating blips or markers to show when agent i switches from optimizing personal utility to status within these graphs, and vice versa.

The most important graph would be the graph of the follower progression over time; the marked points where an agent switches to optimizing for status (when under a certain ρ), accompanied by a pause or fall in the number of followers the agent i receives, would show the effects of threshold clearly.

9.5 Variance

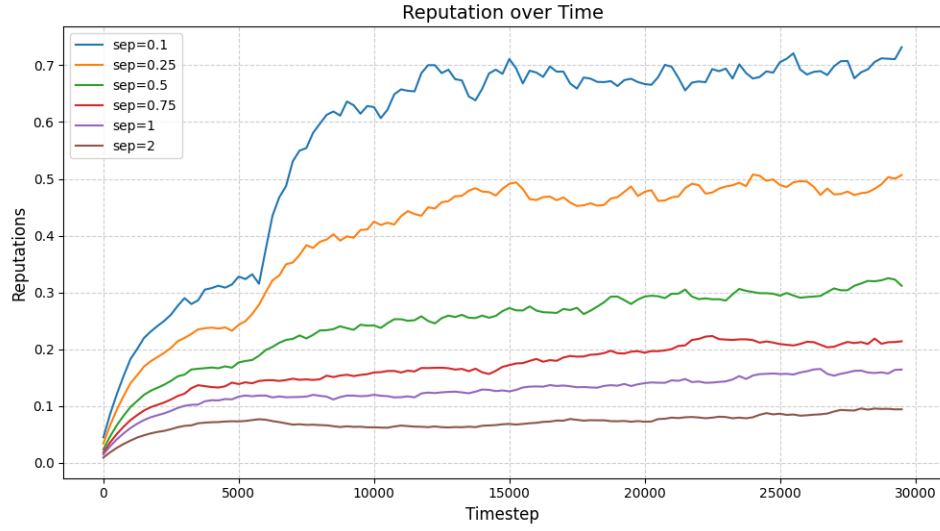


Figure 27: Reputations for $\kappa = 0$ and unimodal varying σ .

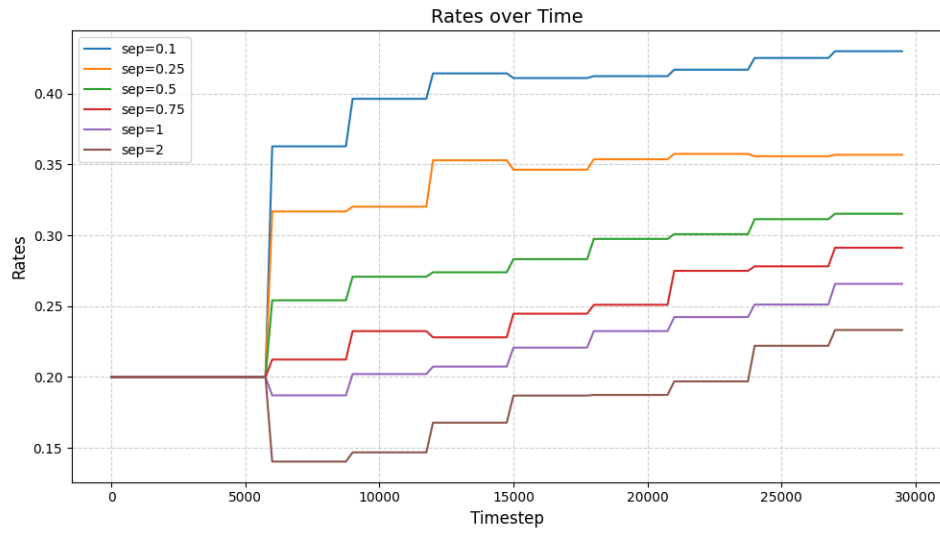


Figure 28: Rates for $\kappa = 0$ and unimodal varying σ .

9.6 Basic Gossiping Rate Changes

9.6.1 Fully Connected

9.6.2 Singularly Connected

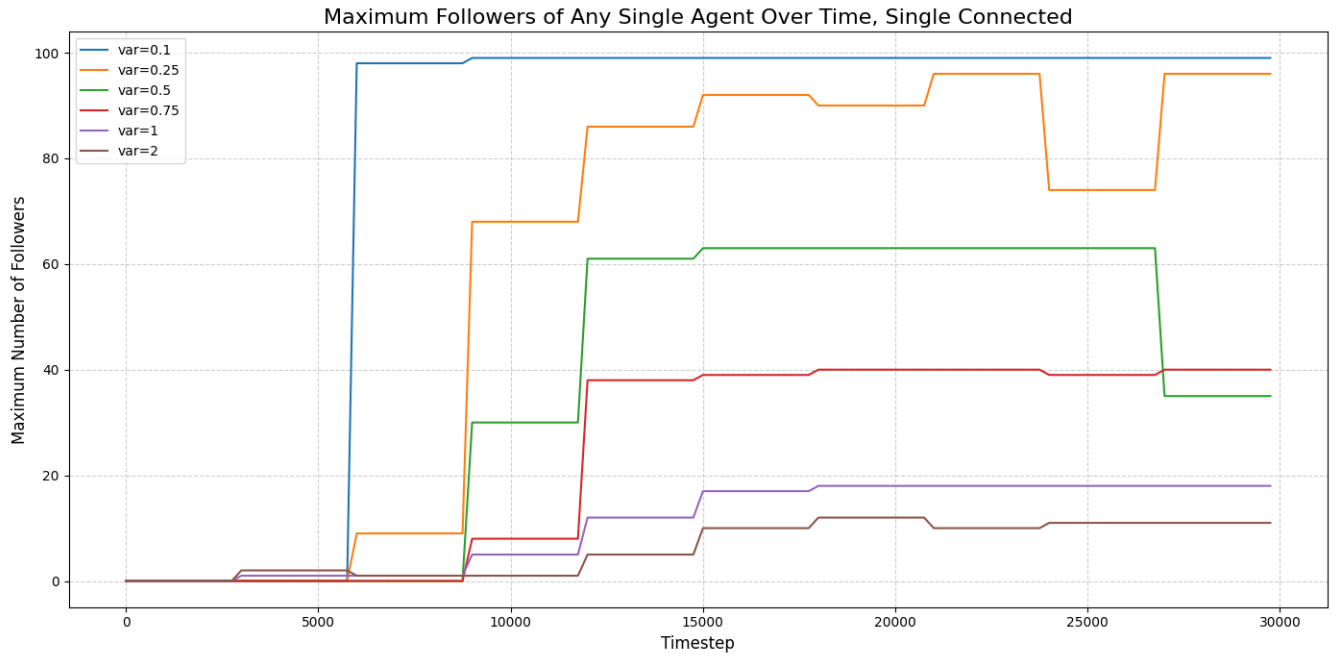


Figure 29: Total followers for $\kappa = 0$ and unimodal varying σ , synchronous

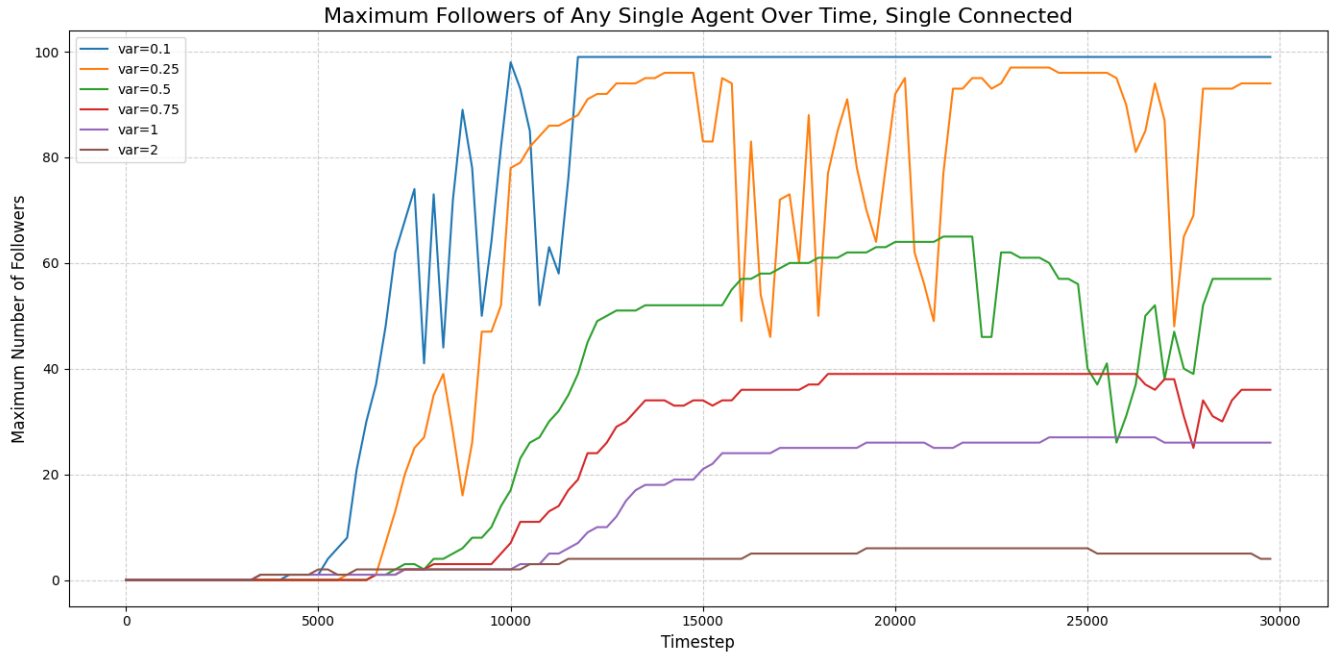


Figure 30: Total followers for $\kappa = 0$ and unimodal varying σ , asynchronous

9.7 Dynamic Binary

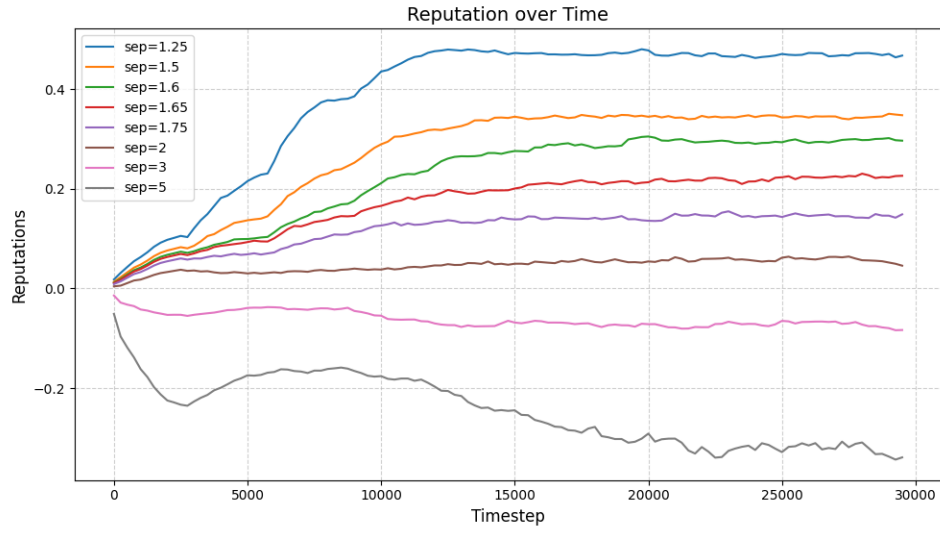


Figure 31: Reputations for $\kappa = 0.1$ and bimodal varying separation.

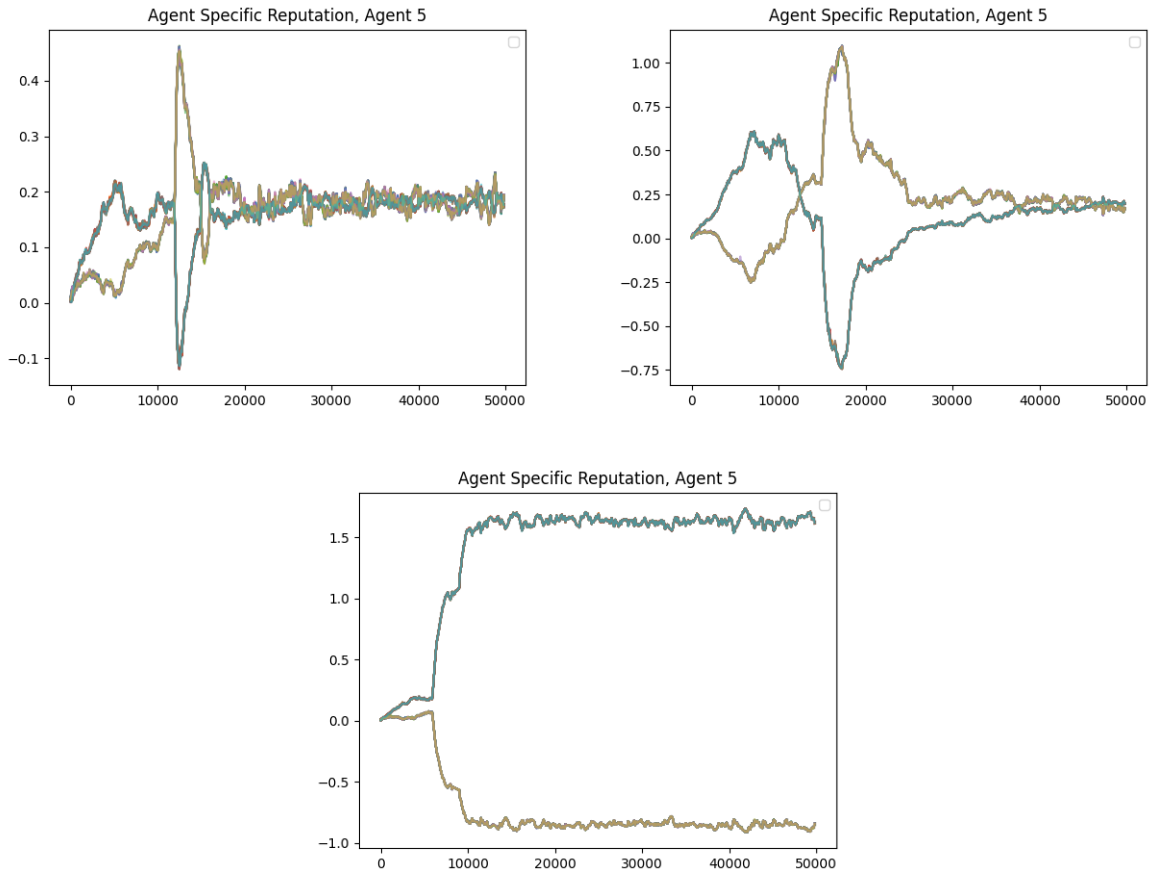


Figure 32: Reputations as seen by Agent 5 for the singularly connected network. Notice how we can see the reputations of the two polarized parts of the community converge faster.



Figure 33: Reputations across the entire community for the Dynamic Binary Connections

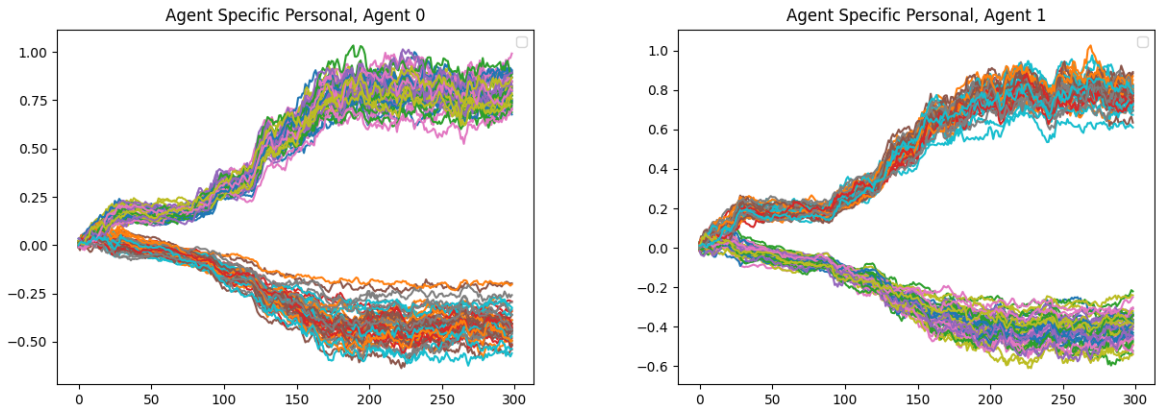


Figure 34: The personal utility that is provided to Agent 0 (left) and Agent 1 (right) by all agents. Note that this is two distinct communities following very different policies.

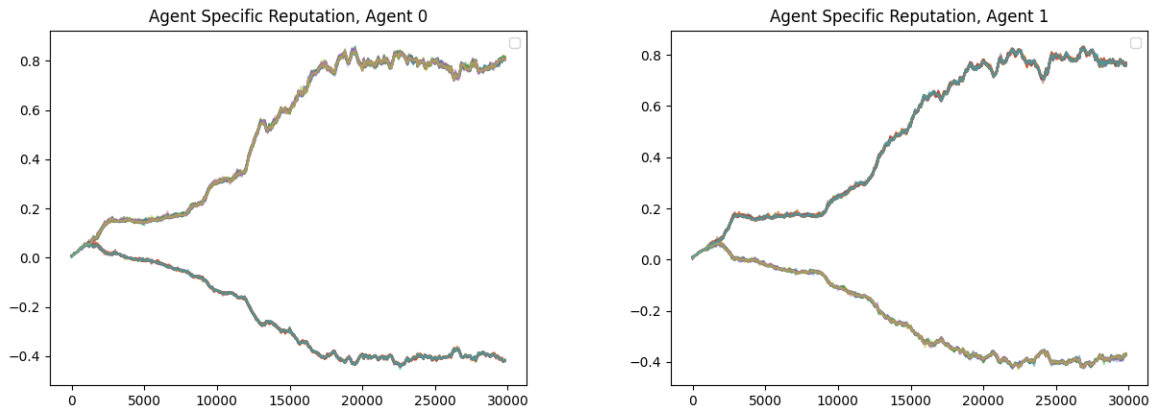


Figure 35: The reputation of Agent 0 (left) and Agent 1 (right) as seen by all agents.

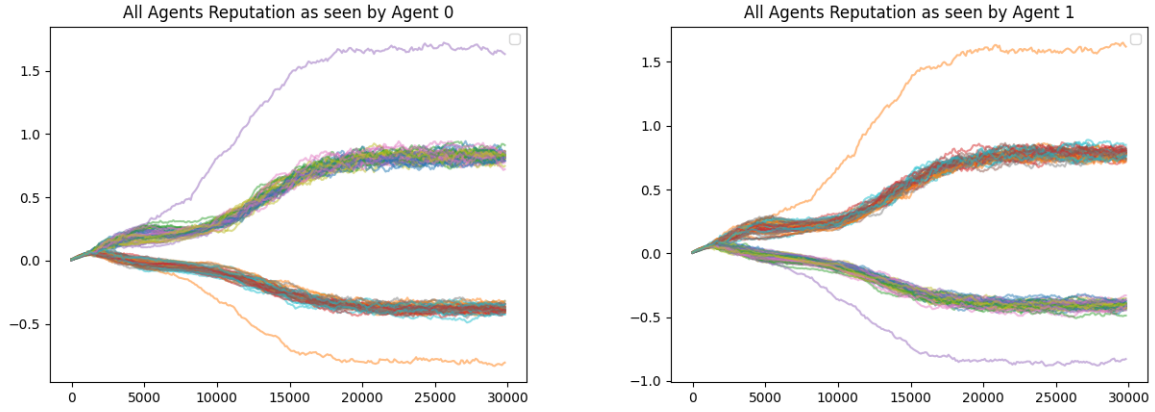


Figure 36: The reputations of all agents as seen by Agent 0 (left) and Agent 1 (right).

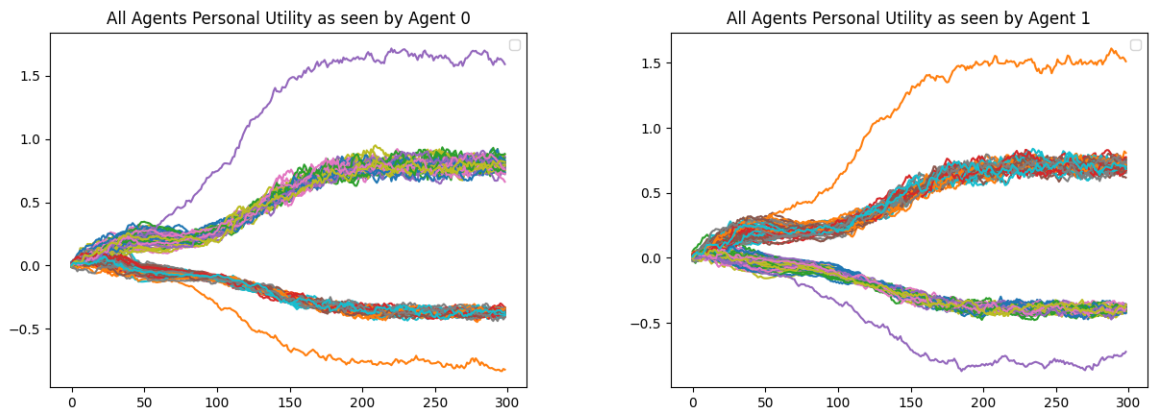


Figure 37: The personal utility provided by all agents to Agent 0 (left) and Agent 1 (right).